MOTION IN A STRAIGHT LINE 15

2.1 Introduction 15

An object moving along a straight line has position x at time t. Displacement is the change in position. Velocity is the rate of change of position. Acceleration is the rate of change of velocity.

For uniform acceleration the equations are v = u + a*t and s = u*t + (1/2)*a*t*t. A car starting from rest gains speed uniformly; its average velocity is (u + v)/2.